

1) Vectorization and Embedding Techniques

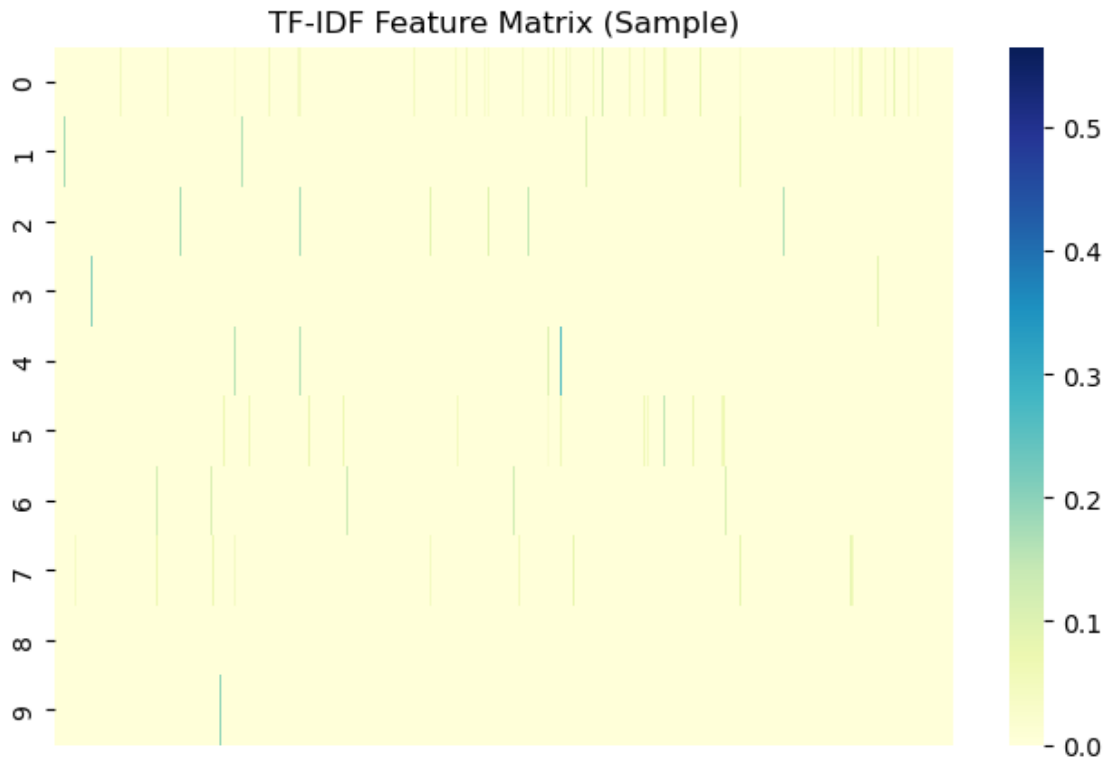
i. TF-IDF Vectorization

Term Frequency–Inverse Document Frequency (TF-IDF) vectorization was applied to the preprocessed review texts to convert the corpus into a structured numeric feature matrix suitable for machine learning.

- **Preprocessing:** The reviews were tokenized, lemmatized, and joined into strings of tokens.
- **Vectorization parameters:**
 - N-grams: unigrams and bigrams (ngram_range=(1, 2))
 - Vocabulary size was limited to 5000 most frequent n-grams (max_features=5000)
 - Words appearing in fewer than 5 documents were excluded (min_df=5)
 - Words appearing in more than 90% of documents were also ignored (max_df=0.9)
- **Result:**

The resulting TF-IDF matrix has dimensions (number_of_reviews × number_of_features). It is sparse, as expected, with many zero entries due to the nature of text data. The sparsity was quantified to assess the density of meaningful features.
- **Visualization:**

A heatmap of a sample of the TF-IDF matrix was generated to visually inspect the distribution of term importance across the sample reviews.



The heatmap visualizes a sample of the TF-IDF feature matrix, where rows represent individual text samples and columns represent terms. Most values are near zero, highlighting the sparse nature of TF-IDF representations. Brighter cells indicate terms with higher TF-IDF scores, meaning they are more informative and distinctive within that specific text.

ii. Word Embeddings (Word2Vec)

To capture semantic relationships and contextual meaning between words beyond simple frequency counts, Word2Vec embeddings were employed.

- **Model training:**

A Skip-gram Word2Vec model was trained on the corpus of tokenized reviews. Alternatively, a previously saved model was loaded to save computation.

- **Embedding generation:**

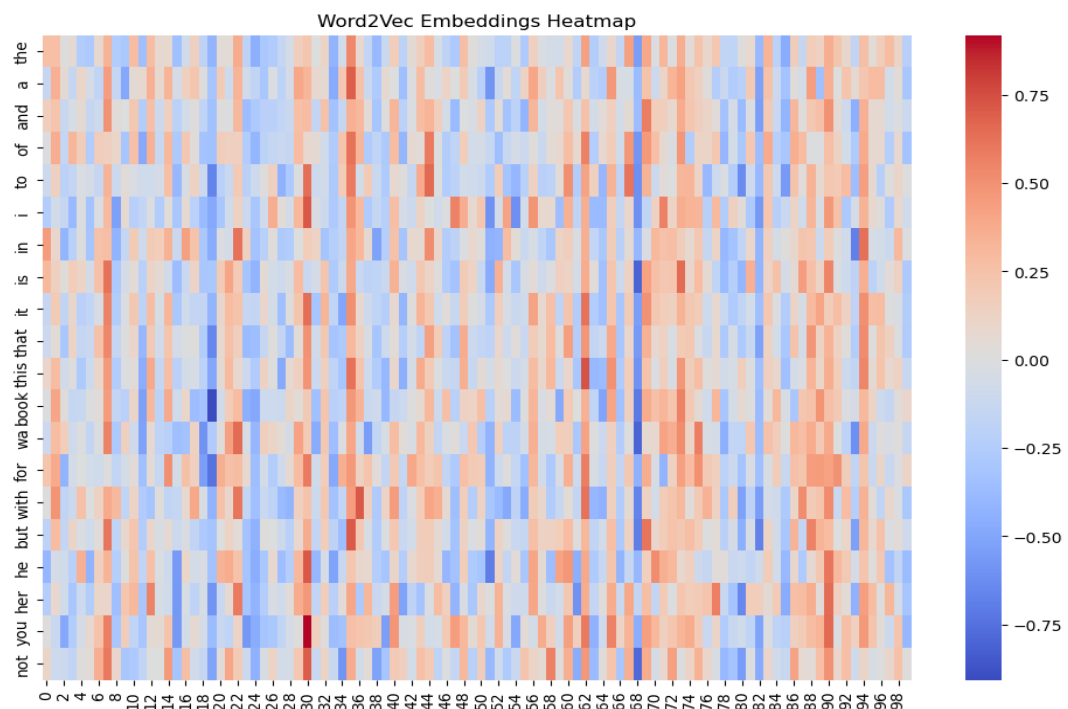
Each review was transformed into a fixed-length dense vector by averaging the Word2Vec vectors of its constituent words. This aggregation method produces a meaningful vector representation of the entire review while preserving semantic richness.

- **Model parameters:**

- Vector size: 100 dimensions
- Context window: 5
- Minimum word count: 2 (words occurring less than twice were ignored)

- **Visualization:**

Heatmaps of selected word vectors were generated to illustrate the variability and semantic patterns in the learned embeddings.



This heatmap visualizes Word2Vec word embeddings, where each row corresponds to a word and each column to a dimension in the embedding space. The color intensity represents the value of each dimension, with red indicating positive values and blue indicating negative ones, helping to reveal similarities and variations in how different words are represented semantically.

Word2Vec embeddings provide dense, low-dimensional representations capturing semantic similarity, which are well suited for neural models or other algorithms benefiting from semantic features.

2) TF-IDF Configuration and Parameter Tuning

```
TfidfVectorizer(  
    ngram_range=(1, 2),  
    max_features=5000,  
    min_df=5,  
    max_df=0.7,  
)
```

Parameter	Value	Explanation
ngram_range	(1, 2)	Includes both unigrams (single words) and bigrams (pairs)
max_features	5000	Limits the feature set to the 5,000 most important n-grams
min_df	5	Discards n-grams appearing in fewer than 5 documents
max_df	0.9	Discards n-grams appearing in more than 90% of documents

Impact of these parameters:

- **N-gram range (1, 2):** Captures richer linguistic patterns by including common two-word phrases (bigrams), which may carry more context than unigrams alone.
- **min_df=5:** Filters out rare terms to reduce noise and overfitting.
- **max_df=0.9:** Keeps terms that aren't too common across almost all documents, thus filtering out uninformative features.
- **max_features=5000:** Controls dimensionality for computational efficiency and to avoid sparse features explosion.

3) Comparison: TF-IDF vs Word Embeddings (Dimensionality & Interpretability)

Aspect	TF-IDF	Word2Vec
Type	Sparse, frequency-based representation	Dense, learned vector representation

Vector Size	Determined by vocabulary size (e.g., 5000 features in your config)	Fixed size (e.g., 100 dimensions, as set in your model)
Representation Basis	Counts and inverse frequency of words across documents	Semantic similarity from surrounding word contexts
Context Awareness	No – treats each word independently	Yes – captures word meaning from surrounding context
Interpretability	High – each feature corresponds to an actual word or n-gram	Low – vectors capture meaning implicitly, not human-readable
Sparsity	High – most entries in the matrix are zeros	Low – compact and continuous vectors
Training Requirement	None – uses predefined formula	Requires training or pre-trained models
Handling Synonyms	Poor – treats synonyms as unrelated tokens	Good – maps semantically similar words to nearby vectors
Use Cases	Document classification, keyword extraction, information retrieval	Clustering, similarity matching, sentiment analysis, deep learning input
Downstream Usage	Works well with linear models and traditional ML	Better suited for deep learning and non-linear models

4) Sentiment Labeling Logic

To enable supervised learning or evaluation, sentiment labels were assigned to reviews using the rule-based VADER sentiment analysis tool:

- **Scoring:**

VADER computes a compound polarity score between -1 (most negative) and +1 (most positive) for each review.

- **Labeling thresholds:**

- Compound score $\geq 0.05 \rightarrow$ Positive sentiment
- Compound score $\leq -0.05 \rightarrow$ Negative sentiment
- Scores between -0.05 and 0.05 $\rightarrow$ Neutral sentiment

- **Labeling Criteria:**

if compound ≥ 0.05 :

 sentiment = 'positive'

elif compound ≤ -0.05 :

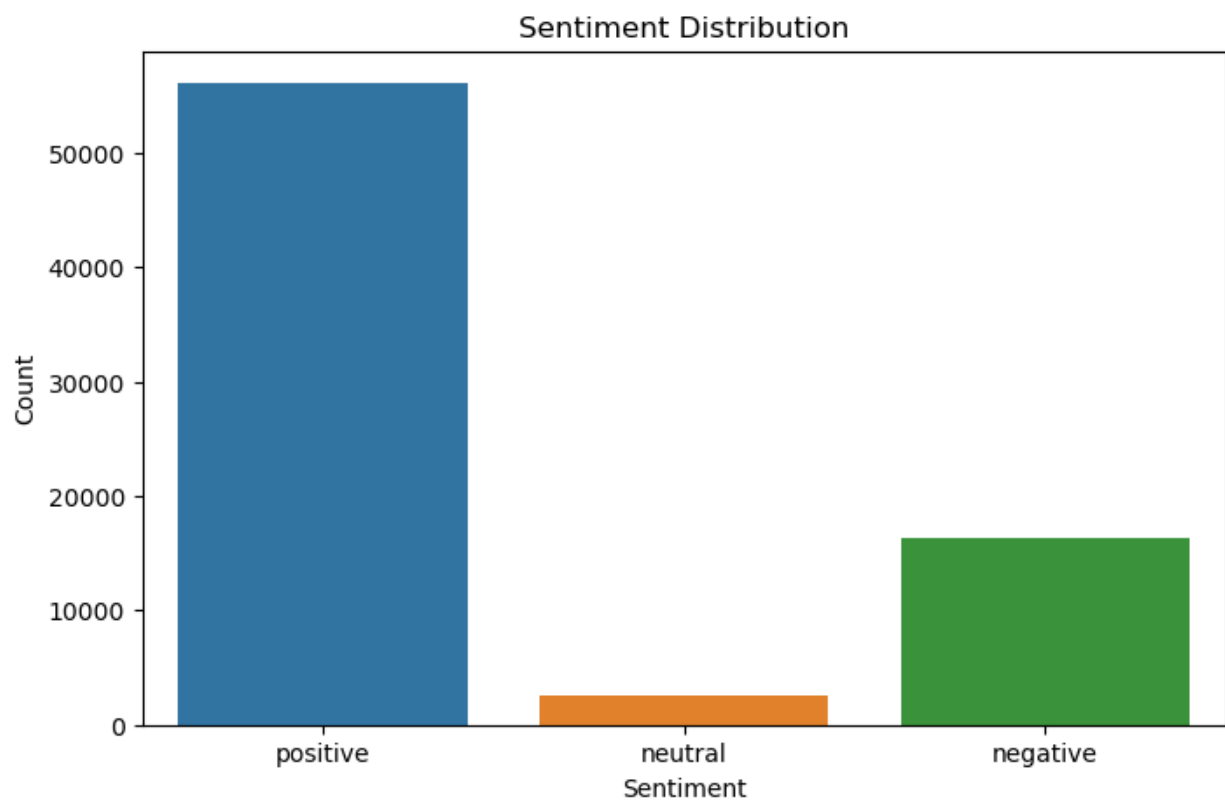
 sentiment = 'negative'

else:

 sentiment = 'neutral'

- **Distribution:**

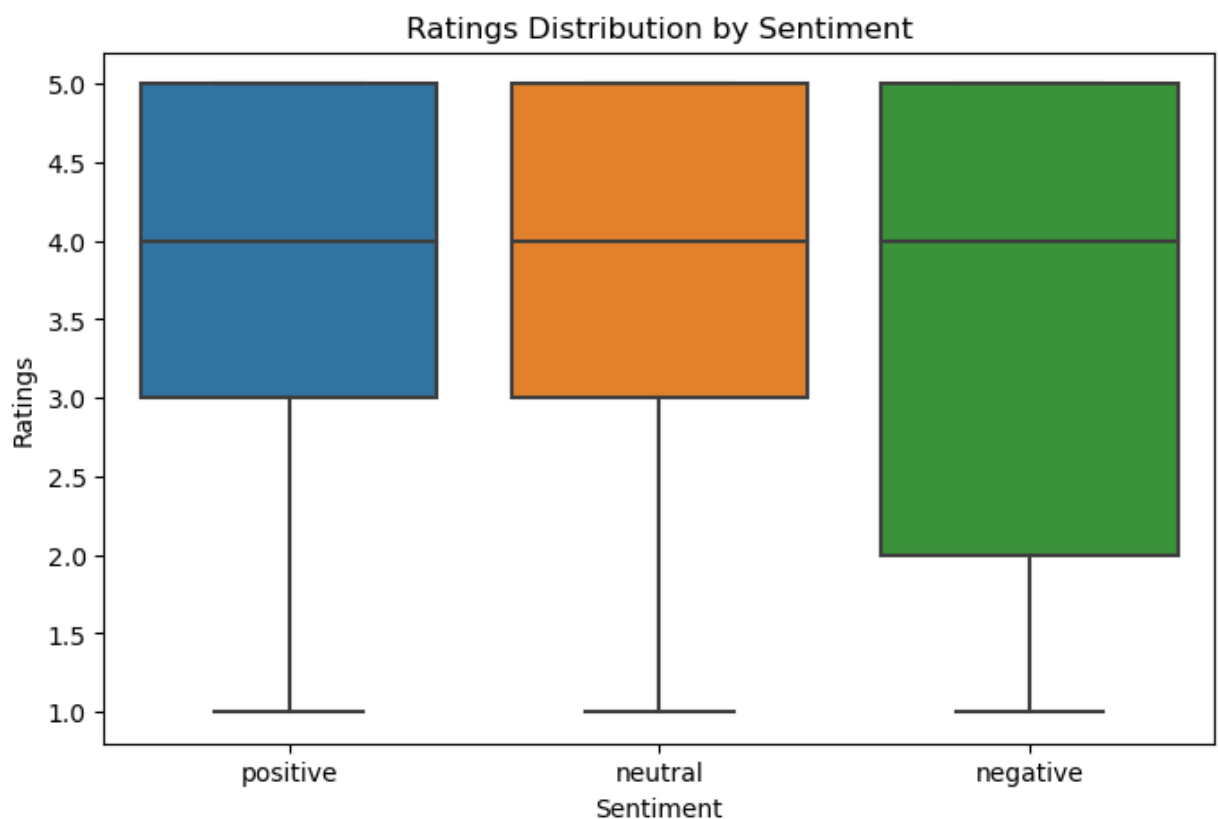
The frequency of positive, neutral, and negative labels was analyzed and visualized, showing the sentiment balance in the dataset.



The bar chart shows the distribution of sentiment labels in the dataset, with positive reviews dominating the sample. Negative sentiments are present in moderate numbers, while neutral sentiments are significantly underrepresented.

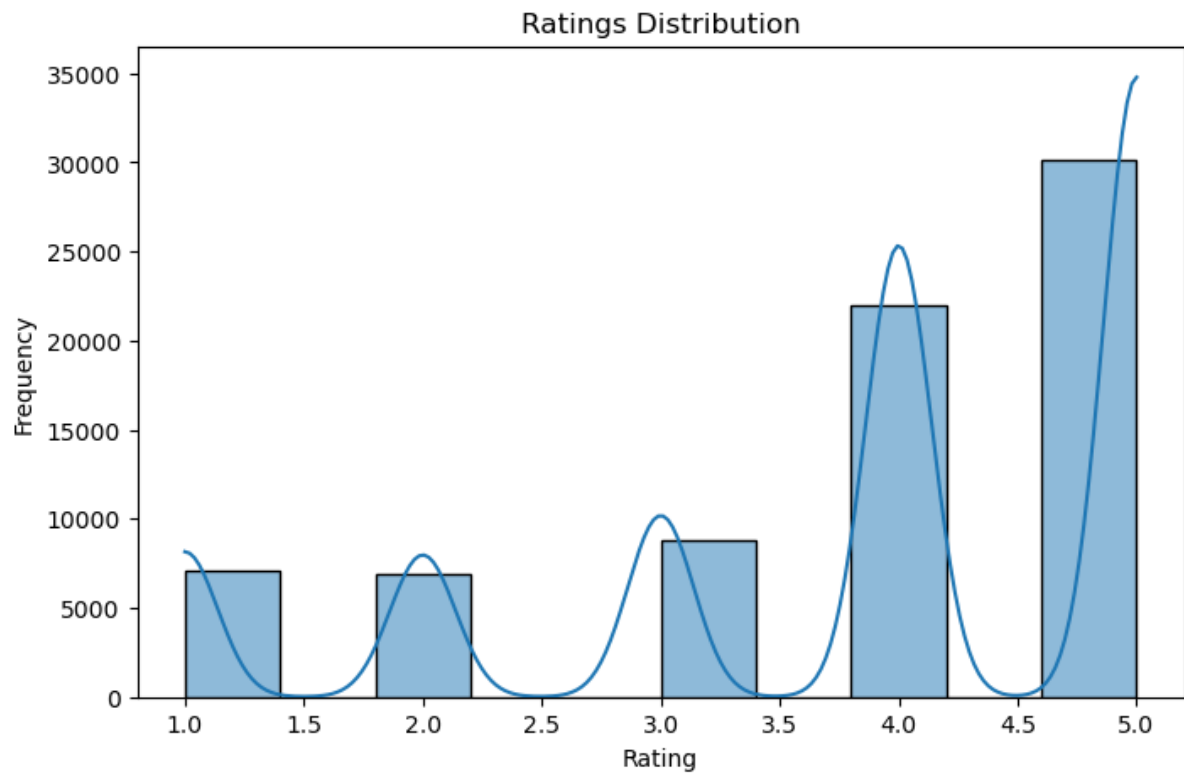
- **Additional analysis:**

Ratings distribution was examined by sentiment category to validate the consistency of sentiment labeling with user ratings.



The box plot displays the distribution of ratings across sentiment categories. Despite the differing sentiments, the median rating remains consistent at 4 for all three groups, but negative sentiments show a wider spread, with many ratings falling as low as 1, indicating a broader variance in how users rate negatively worded reviews.

Ratings Distribution



The histogram with KDE curve illustrates the distribution of ratings across the dataset. The majority of the ratings are skewed towards 5 stars, indicating a generally positive sentiment. Ratings of 4 also have a high frequency, while lower ratings (1 to 3) occur much less often, suggesting that most users provided favorable reviews.